SRM TRP Engineering College
Tiruchirappalli
Course Assignment Instructions with Rubrics

Department	B.Tech AI&DS							
Programme:			**Semester:**	V	**AY:**	III		
Faculty Handling:	R. KALAIYARASI							
Course Code & Title:	AD3501 & DEEP LEARNING							
Assessment Task:			**Assessment Type:**	Term Work/Self			**Required Hrs:**	
Assessment Question based on:	*Open Ended*	✓	*Problem Based Learning*		✓	*Creative Thinking*		
	Close Ended		*Project Based Learning*			*Critical Thinking*		✓
	Any Other Type	MCQ or any other	*Challenge Based Learning*		✓	*Level of Thinking (RBT)*		K5
SDG to be Addressed:	SDG is Required	Yes/No	*SDG Number (Specify or Mention Students Choice)*			*Target Number (If Applicable)*		
SDG Integration Level: (If Addressed)	*Level 1: Awareness*		*Level 2: Design Alignment*		✓	*Level 3: Impact and Measurement*		
RIASEC Context	Realistic	Investigative ✓	Artistic	Social		Enterprising	Conventional ✓	
Presentation Format:	PowerPoint/Poster/Video/PDF/Written/Any other (Specify)							
Max Marks:	Scaled 38							
Relevant Course Outcome(s):								
Submission Deadline:								

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Instructions to the Students:
Assessment Question: Emotion Recognition from facial images using Deep Learning
Guidelines: E.g. Guidelines for Preparation/Submission/Expected Output/Review Process/Presentation Format if any

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Rubrics for Evaluation:

Performance Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)

Team Members:

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30/40

R. Sahay

AI ~~pdf~~ Content

AD3501 – DEEP LEARNING

Team Literature Survey Report

Topic:

Emotion Recognition from Facial Images using Deep Learning

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1.1 Combined Literature Survey Comparison Table

The table below consolidates all ten reviewed papers into a single comparison view — title and year, authors, journal quartile, methodology, dataset(s), model, headline result, and key limitation — providing a quick reference before the detailed research-gap discussion that follows in Section 2.

S. No	Paper Title (Year)	Author(s)	Journal (Quartile)	Methodology	Dataset(s)	Model	Result	Limitations
1	A Comprehensive Deep Learning Framework for Real-Time Emotion Detection in Online Learning Using Hybrid Models (2025)	M. Aly, N. S. Alotaibi	Scientific Reports (Q1)	New annotated dataset; fine-tuned backbone with attention; temporal modeling; optimization for real-time FER	FER2013, CK+, KDEF, own dataset	ResNet-50 + CBAM + 3D CNN + AGTO	95.57% (FER2013); 97.29% (CK+); 98.35% (KDEF); 98.09% (own dataset)	Heavy stacked components — real-time computational cost undiscussed; partial testing on an unpublished own dataset limits verifiability.
2	Application of Multiple Deep Learning Architectures for Emotion Classification Based on Facial Expressions (2025)	C. Qian, J. A. Lobo Marques, A. R. de Alexandria, S. Fong	Sensors (Q1)	Comparative evaluation of 10 DL architectures; accuracy, training time, and model-weight-size analysis	FER2013	VGG16, VGG19, ResNet50, ResNet101, DenseNet, GoogLeNet, MobileNet, EfficientNetV2, ShuffleNetV2, RepVGG	EfficientNetV2 best at 68.7%; ResNet50/101 close behind at 68.1%	Purely comparative — no new architecture proposed; single-dataset evaluation; accuracy ceiling under 69%.
3	A Novel Emotion Recognition System for Human–Robot Interaction (HRI) Using Deep Ensemble Classification (2025)	K. Zaman et al.	Int. J. Intelligent Systems (Q1)	Face detection in video; transfer learning; pooled features from four CNNs; ensemble classification	CK+, FER-2013, HERD	EfficientNet-TL ensemble with Faster R-CNN and VGG-19 block	89.23% (CK+); 94.36% (FER-2013); 97.01% (HERD)	Four-CNN ensemble likely adds latency, unaddressed for real-time HRI; accuracy varies widely (89–97%) across datasets.
4	Emotion Recognition in Autistic Children Through Facial Expressions Using Advanced Deep Learning Architectures (2025)	P. Radocaj, G. Martinovic	Applied Sciences (Q2)	Direct comparison of CNN vs. transformer models on a four-class emotion task specific to ASD children	Labeled ASD facial image dataset	Eight CNN/transformer models compared; best = Swin Transformer	Swin Transformer accuracy 0.8000; F1 0.7889	Small, population-specific ASD dataset; results may not generalize to larger, more diverse groups.
5	A Generalized Zero-Shot Deep Learning Classifier for Emotion Recognition Using Facial Expression Images (2025)	V. Bhati, N. Tiwari, M. Chawla	IEEE Access (Q1)	Zero-shot cross-dataset generalization testing on unseen expression classes	FER2013, AffectNet, RAF-DB, CK+, KDEF, JAFFE	GZS-ConvNet	Reported as highly generalizable across all six benchmark datasets	No class-wise accuracy for unseen categories; accuracy-vs-flexibility trade-off not clearly measured.
6	Emotion Recognition for Enhanced Learning: Using AI to Detect Students' Emotions and Adjust Teaching Methods (2025)	S. A. Salloum et al.	Smart Learning Environments (Q1)	FER2013 preprocessing; data augmentation; 80:20 split; early stopping; custom CNN	FER2013	Refined CNN with batch normalization and dropout	95% test accuracy across seven emotion classes	Tested only on FER2013, not real classroom footage; the "adjust teaching methods" claim is unproven in practice.

S. No	Paper Title (Year)	Author(s)	Journal (Quartile)	Methodology	Dataset(s)	Model	Result	Limitations
7	A Novel Transformer-Based Approach for Adult's Facial Emotion Recognition (2025)	U. Nawaz, Z. Saeed, K. Atif	IEEE Access (Q1)	Attention mechanism; dynamic token pruning; multi-dataset evaluation; ablation studies	FER2013, CK+, AffectNet, RAF-DB, AFEW	Transformer-based FER model	98.67% (FER2013); 99.52% (CK+); 99.3% (AffectNet); 98.45% (RAF-DB); 96.3% (AFEW)	Near-99% accuracy across datasets raises overfitting/leakage concerns; extra compute from token pruning not quantified.
8	Attention-Enhanced Optimized Deep Ensemble Network for Effective Facial Emotion Recognition (2025)	T. Khan, M. Yasir, C. Choi	Alexandria Engineering Journal (Q1)	Augmentation; super-resolution; parallel backbone ensemble; channel and spatial attention	FER, KDEF	EA-Net: EfficientNetB0 + InceptionV3 + CAM + SAM	78.60% (FER); 99.30% (KDEF)	Large accuracy gap between FER and KDEF shows weak performance under uncontrolled conditions; pipeline is heavy for lightweight deployment.
9	Enhancing Facial Expression Recognition in Uncontrolled Environment: A Lightweight CNN Approach with Pre-processing (2025)	R. Grover, S. Bansal	Neural Computing and Applications (Q1)	Lightweight CNN evaluated with and without preprocessing, under controlled and uncontrolled settings	FER-2013, CK+	LCNN (Lightweight CNN)	65% (FER-2013); 98% (CK+)	FER-2013 accuracy only 65%, showing a real efficiency-accuracy trade-off; "uncontrolled" testing not on truly varied wild images.
10	A Fine-Grained Human Facial Key Feature Extraction and Fusion Method for Emotion Recognition (2025)	S. Li et al.	Scientific Reports (Q1)	Super-resolution and lighting preprocessing; global-local feature extraction; adaptive fusion	FER-2013, JAFFE, self-built dataset	Two-branch CNN with adaptive fusion	80.59% (FER-2013); 97.61% (JAFFE)	Uses a non-public self-built dataset, limiting reproducibility; complex two-branch fusion gives only modest gain over simpler CNNs.

Note: Journal quartiles reflect the SJR best-quartile ranking (SCImago Journal & Country Rank, most recent release) as of Aug 2026; please re-verify against your institution's approved source before final submission.

2. Research Gap Analysis of Each Paper

Each subsection below identifies the specific research gap left open by the corresponding paper — i.e., the concrete, unresolved problem that limits the applicability, reproducibility, or generalizability of the reported approach, and which justifies further investigation.

2.1 Paper 1: A Comprehensive Deep Learning Framework for Real-Time Emotion Detection in Online Learning Using Hybrid Models (2025)

Authors: *M. Aly and N. S. Alotaibi — Scientific Reports (Q1)*

Identified Research Gap:

The paper stacks ResNet-50, a 3D CNN, CBAM attention, and an AGTO optimizer into a single pipeline but never reports inference latency, FLOPs, or memory footprint, so the claim of "real-time" performance on ordinary classroom hardware remains unverified. A meaningful fraction of the reported gains (98.09% on the "own dataset") also rests on a privately held dataset, which prevents independent replication and leaves the true generalization ability of the hybrid model an open question.

2.2 Paper 2: Application of Multiple Deep Learning Architectures for Emotion Classification Based on Facial Expressions (2025)

Authors: *C. Qian, J. A. Lobo Marques, A. R. de Alexandria, and S. Fong — Sensors (Q1)*

Identified Research Gap:

Because the study is purely a comparative benchmark of ten existing backbones, it contributes no new modelling idea to push accuracy beyond the 68–69% ceiling observed on FER2013. Evaluation is restricted to a single, fairly noisy dataset, so the relative rankings of the ten architectures (EfficientNetV2, ResNet variants, MobileNet, etc.) cannot be assumed to hold on cleaner or more diverse corpora, leaving cross-dataset architecture comparison as an unresolved gap.

2.3 Paper 3: A Novel Emotion Recognition System for Human–Robot Interaction (HRI) Using Deep Ensemble Classification (2025)

Authors: *K. Zaman et al. — International Journal of Intelligent Systems (Q1)*

Identified Research Gap:

Pooling features from four separate CNNs (Faster R-CNN, VGG-19, and an EfficientNet-based ensemble) is likely to introduce non-trivial inference latency, yet the paper never measures response time — a critical omission for HRI, where a robot must react to a human's expression in real time. The accuracy swing from 89.23% (CK+) to 97.01% (HERD) further indicates the ensemble is dataset-dependent rather than genuinely robust across recording conditions, sensors, and camera angles typical of robotic platforms.

2.4 Paper 4: Emotion Recognition in Autistic Children Through Facial Expressions Using Advanced Deep Learning Architectures (2025)

Authors: *P. Radocaj and G. Martinovic — Applied Sciences (Q2)*

Identified Research Gap:

The study benchmarks eight generic CNN/transformer models on a small, population-specific dataset of autistic children's expressions without proposing any architectural adaptation tailored to the atypical, often more subtle facial affect displayed by this group. With accuracy capped at 80%, and no discussion of dataset size, class balance, or demographic diversity, the work leaves open whether the reported gains would hold on a larger, more representative ASD population or generalize across age groups and cultural backgrounds.

2.5 Paper 5: A Generalized Zero-Shot Deep Learning Classifier for Emotion Recognition Using Facial Expression Images (2025)

Authors: *V. Bhati, N. Tiwari, and M. Chawla — IEEE Access (Q1)*

Identified Research Gap:

Although the GZS-ConvNet targets the practically important zero-shot setting, the paper reports only aggregate generalization claims across six benchmarks and omits class-wise accuracy for the unseen emotion categories, making it impossible to assess where the model actually fails. There is also no quantified trade-off between recognition accuracy and generalization flexibility, so it is unclear how much accuracy is sacrificed to achieve zero-shot capability — a gap that limits the method's practical adoption.

2.6 Paper 6: Emotion Recognition for Enhanced Learning: Using AI to Detect Students' Emotions and Adjust Teaching Methods (2025)

Authors: *S. A. Salloum et al. — Smart Learning Environments (Q1)*

Identified Research Gap:

Despite framing itself as a tool to "adjust teaching methods," the system is validated only on the static FER2013 image set and never tested against live classroom video, variable lighting, partial occlusion (masks, hands, notebooks), or multiple simultaneous students. The pedagogical adaptation loop — detecting an emotion and then actually changing the teaching strategy — is proposed conceptually but not implemented or evaluated, leaving the core educational claim of the paper empirically unverified.

2.7 Paper 7: A Novel Transformer-Based Approach for Adult's Facial Emotion Recognition (2025)

Authors: *U. Nawaz, Z. Saeed, and K. Atif — IEEE Access (Q1)*

Identified Research Gap:

Accuracies approaching 99% across five different benchmark datasets are unusually high for facial emotion recognition and are not accompanied by a rigorous check for train/test overlap, subject-independent splitting, or data leakage, which raises concerns about overfitting. In addition, the dynamic token-pruning mechanism that gives the transformer its efficiency claim is not benchmarked for the

additional computation it introduces during inference, so the true speed/accuracy trade-off of the approach is not established.

2.8 Paper 8: Attention-Enhanced Optimized Deep Ensemble Network for Effective Facial Emotion Recognition (2025)

Authors: *T. Khan, M. Yasir, and C. Choi — Alexandria Engineering Journal (Q1)*

Identified Research Gap:

EA-Net's accuracy collapses from 99.30% on the controlled KDEF dataset to 78.60% on the more naturalistic FER dataset, revealing that the model has not solved the core challenge of recognizing emotion under uncontrolled, in-the-wild conditions — it merely performs well when lighting and pose are already clean. Fusing two large backbones (EfficientNetB0 and InceptionV3) with channel and spatial attention and a super-resolution stage also produces a computationally heavy pipeline that is left unexamined for deployment on lightweight or embedded devices.

2.9 Paper 9: Enhancing Facial Expression Recognition in Uncontrolled Environment: A Lightweight CNN Approach with Pre-processing (2025)

Authors: *R. Grover and S. Bansal — Neural Computing and Applications (Q1)*

Identified Research Gap:

While the lightweight CNN is transparent about trading accuracy for efficiency, its 65% accuracy on FER-2013 shows the accuracy loss is still substantial, and the paper's notion of an "uncontrolled environment" is not validated against a genuinely diverse, in-the-wild image set (varied ethnicity, illumination extremes, motion blur, occlusion). This leaves a gap in demonstrating that lightweight models can approach the accuracy of heavier architectures while remaining deployable on resource-constrained devices.

2.10 Paper 10: A Fine-Grained Human Facial Key Feature Extraction and Fusion Method for Emotion Recognition (2025)

Authors: *S. Li et al. — Scientific Reports (Q1)*

Identified Research Gap:

Part of the reported performance depends on a self-built dataset that is not publicly released, which limits independent verification of the fine-grained, two-branch fusion approach. Furthermore, the accuracy gain on FER-2013 (80.59%) over simpler single-branch CNN baselines is modest relative to the additional architectural complexity introduced by the global-local fusion and super-resolution preprocessing stages, suggesting the cost-benefit ratio of the added complexity has not been convincingly demonstrated.

3. Proposed Future Research Directions

Building on the research gaps identified above, the following ten directions are proposed as original, forward-looking contributions this survey suggests for the field of deep-learning-based facial emotion recognition. They are synthesized across the collective evidence of all ten papers rather than tied to any single study, and are intended to guide subsequent project work, thesis proposals, or follow-on research.

1. Latency- and Resource-Aware Real-Time Benchmarking

Future work should report inference latency, FLOPs, parameter count, and memory footprint alongside accuracy for every proposed architecture, especially for models targeting online learning platforms (Papers 1 and 6) and human–robot interaction (Paper 3). A standardized real-time benchmarking protocol — tested on representative low-cost hardware such as a classroom laptop, a Raspberry Pi, or an embedded robotics controller — would let researchers and practitioners compare models on genuinely comparable terms rather than accuracy alone.

2. Population-Specific and Inclusive Model Design

Rather than applying generic CNN/transformer backbones to specialized populations, future research should design architectures and augmentation strategies explicitly informed by the expression characteristics of under-represented groups such as autistic children (Paper 4), elderly users, or individuals with facial differences. This includes collecting larger, ethically sourced, demographically balanced datasets and reporting subgroup-level accuracy rather than a single aggregate score.

3. Robustness to In-the-Wild Conditions

The recurring gap between controlled-dataset accuracy (KDEF, CK+) and uncontrolled-dataset accuracy (FER-2013, FER) observed across Papers 1, 3, 8, and 9 points to a clear need for training strategies — such as adversarial illumination/pose augmentation, domain adaptation, and self-supervised pretraining on unlabeled in-the-wild video — that specifically close this controlled-to-uncontrolled performance gap rather than optimizing for a single benchmark.

4. Explainable and Trustworthy FER for High-Stakes Contexts

For applications in education (Papers 1, 6), autism support (Paper 4), and human–robot interaction (Paper 3), black-box predictions are insufficient. Future systems should integrate explainability techniques (Grad-CAM, attention visualization, concept-based explanations) so that teachers, therapists, or robot operators can understand and trust why a particular emotion label was assigned, and can catch systematic misclassifications before they affect real decisions.

5. Rigorous Zero-Shot and Cross-Dataset Evaluation Protocols

Building on the zero-shot direction of Paper 5, future studies should adopt subject-independent, class-wise reporting for unseen-category performance and explicitly quantify the accuracy-versus-generalization trade-off, rather than reporting only aggregate cross-dataset success. Leave-one-dataset-out and leave-one-class-out protocols should become standard practice for claims of generalization.

6. Open, Reproducible, and Standardized Datasets

Several papers (1, 6, 10) rely partly or wholly on privately held or self-built datasets, undermining reproducibility. The field would benefit from community-driven, IRB-approved, publicly released benchmark datasets that specifically capture uncontrolled classroom, robotic, and assistive-technology recording conditions, with clear licensing for academic reuse.

7. Overfitting-Aware Reporting for Near-Perfect Accuracy Claims

Given the unusually high accuracies reported in Paper 7 (up to 99.52%), future transformer-based FER research should mandate subject-independent train/test splitting, k-fold cross-validation, and explicit leakage checks, so that near-ceiling performance can be trusted rather than viewed with suspicion.

8. Multimodal and Temporal Fusion

Static single-frame image classification, used by most of the surveyed papers, misses temporal dynamics and complementary cues. Combining facial video sequences with voice tone, physiological signals (heart rate, EEG where available), and body posture could substantially improve robustness, particularly for ambiguous or low-intensity expressions that single-modality FER frequently misclassifies.

9. Lightweight-Yet-Accurate Architectures for Edge Deployment

The accuracy-efficiency trade-off highlighted in Paper 9 (65% for a lightweight CNN) suggests an open opportunity for knowledge distillation, neural architecture search, and quantization-aware training to shrink heavy ensemble models such as those in Papers 1, 3, and 8 down to edge-deployable sizes without the current double-digit accuracy penalty.

10. Real-World Field Validation Beyond Static Benchmarks

Almost every surveyed paper validates exclusively on curated image datasets. A important future direction is longitudinal field deployment — in real classrooms, robotics labs, or clinical settings — with human-rated ground truth, to determine whether laboratory accuracy actually translates into a system that meaningfully improves teaching, therapy, or interaction outcomes.

4. Sustainable Development Goal (SDG) Mapping

Facial emotion recognition research, while primarily a technical deep-learning discipline, has direct downstream relevance to several United Nations Sustainable Development Goals through its applications in education, health and well-being, inclusive technology design, and industrial innovation. The mapping below connects the surveyed literature to the most relevant SDGs.

SDG	Relevance to Emotion Recognition Research
SDG 3 — Good Health and Well-Being	Emotion-aware systems support mental-health screening and behavioural monitoring; Paper 4's work on recognizing emotions in autistic children directly supports early identification of distress and communication difficulty, aiding therapeutic and clinical intervention.
SDG 4 — Quality Education	Papers 1 and 6 apply facial emotion recognition to online and classroom learning, aiming to detect student engagement, confusion, or frustration in real time so that teaching methods can be adapted — directly advancing inclusive and effective learning environments.
SDG 8 — Decent Work and Economic Growth	Emotion-recognition-enabled human–robot interaction (Paper 3) supports safer, more responsive collaborative robots in workplaces and service industries, contributing to productivity gains and improved human-machine collaboration.
SDG 9 — Industry, Innovation and Infrastructure	The comparative architecture studies (Papers 2, 7, 8, 9, 10) drive innovation in efficient, resilient deep learning infrastructure — a foundation for embedding affective computing into consumer devices, robotics, and smart-education infrastructure.
SDG 10 — Reduced Inequalities	Population-specific and zero-shot generalization research (Papers 4 and 5) works toward emotion-recognition systems that perform equitably across neurodiverse individuals, age groups, and previously unseen expression categories, reducing algorithmic bias against under-represented users.
SDG 17 — Partnerships for the Goals	The reproducibility gaps identified across several papers (reliance on private/self-built datasets) highlight the need for open data-sharing partnerships between universities, hospitals, and technology companies to build larger, public, ethically-sourced emotion datasets for the wider research community.

Taken together, the surveyed papers show that the strongest SDG alignment of current FER research lies in education (SDG 4) and inclusive, equitable technology design (SDG 10), while the weakest alignment — and therefore the greatest opportunity for future contribution — lies in open-data partnerships (SDG 17), since half of the reviewed studies depend at least partly on private or self-built datasets that restrict wider community benefit.

6. References (IEEE Format)

[1] M. Aly and N. S. Alotaibi, "A comprehensive deep learning framework for real-time emotion detection in online learning using hybrid models," Sci. Rep., vol. 15, 2025.

[2] C. Qian, J. A. Lobo Marques, A. R. de Alexandria, and S. Fong, "Application of multiple deep learning architectures for emotion classification based on facial expressions," Sensors, vol. 25, 2025.

[3] K. Zaman et al., "A novel emotion recognition system for human–robot interaction (HRI) using deep ensemble classification," Int. J. Intell. Syst., 2025.

[4] P. Radocaj and G. Martinovic, "Emotion recognition in autistic children through facial expressions using advanced deep learning architectures," Appl. Sci., vol. 15, 2025.

[5] V. Bhati, N. Tiwari, and M. Chawla, "A generalized zero-shot deep learning classifier for emotion recognition using facial expression images," IEEE Access, vol. 13, 2025.

[6] S. A. Salloum et al., "Emotion recognition for enhanced learning: Using AI to detect students' emotions and adjust teaching methods," Smart Learn. Environ., vol. 12, 2025.

[7] U. Nawaz, Z. Saeed, and K. Atif, "A novel transformer-based approach for adult's facial emotion recognition," IEEE Access, vol. 13, 2025.

[8] T. Khan, M. Yasir, and C. Choi, "Attention-enhanced optimized deep ensemble network for effective facial emotion recognition," Alexandria Eng. J., vol. 118, 2025.

[9] R. Grover and S. Bansal, "Enhancing facial expression recognition in uncontrolled environment: A lightweight CNN approach with pre-processing," Neural Comput. Appl., vol. 37, 2025.

[10] S. Li et al., "A fine-grained human facial key feature extraction and fusion method for emotion recognition," Sci. Rep., vol. 15, 2025.